



### General Details

|                   |                      |
|-------------------|----------------------|
| Assessor          | Samuel Prieto Ayllon |
| Assessment Date   | 03/07/2025           |
| Assigned Reviewer | Samuel Prieto Ayllon |
| Next Review Date  | 03/07/2028           |

|                   |                      |
|-------------------|----------------------|
| Signed Off By     | Oraib AlKheetanRojas |
| Signed Off On     | 19/05/2026           |
| Sign Off Comments |                      |

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| Operation Assessed | Submersible ROV – VideoRay Mission Specialist Defender Operation |
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#### Associated with specific area

 Basement 2

 B2 029 (Robotics Lab)

 CTP

#### Description of work area and/or activity assessed

Operation, routine maintenance, and testing of a VideoRay Mission Specialist Defender remotely-operated vehicle (ROV) for research and inspection tasks. Activities include topside set-up (console, power supplies, tether), launch/recovery in water tanks or open water, and piloting with live video, sonar, and manipulator tools. Work is conducted in the lab or field locations with personnel positioned on the deck or poolside and the ROV submerged.

#### Reason for Risk Assessment

To document and control any health, safety, and equipment-damage risks introduced by actively using the Mission Specialist Defender ROV in research operations.

### Overall Current Risk

## High Risk



### Average Number of Persons Affected

2  Employees

2  Students

| Current Rating                                                                                            | Hazard Information                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Task Status                                                                         |
|-----------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| <b>Moderate Risk</b><br> |  <b>Electricity</b><br>The system uses 100-240 VAC shore power (~ 3 kW) and converts to 400 V DC down the tether; subsea batteries supply up to 48 V DC. Damage to cords, wet connectors, or improper grounding could cause electric shock, short-circuit, or fire.                                                                                                                                            |  |
|                                                                                                           | Measures Currently in place to prevent risk of injury                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                     |
|                                                                                                           | <ul style="list-style-type: none"> <li>• Inline GFCI and LIM interlock prevent energising tether until fully mated.</li> <li>• IP-rated console, sealed connectors, and silicone-greased wet-mate plugs.</li> <li>• Visual inspection of mains leads, tether, and battery shells before each dive.</li> <li>• Operator trained to keep power equipment dry and handle power cords only with dry hands.</li> <li>• Fire-rated extinguishers and emergency shutdown button at console.</li> </ul> |                                                                                     |

| Potential Rating                                                                                          | Additional Controls Required                                                                                                                                                                     |
|-----------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Moderate Risk</b><br> | <b>No Remedial Actions Entered</b><br>If there are any reasonably practicable actions which can be taken to reduce the risk associated with this hazard please use the form above to enter them. |

| Current Rating                                                                                        | Hazard Information                                                                                                                                                                                                                                                                                                                           | Task Status                                                                          |
|-------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| <b>High Risk</b><br> |  <b>Moving Parts</b><br>Seven brushless thrusters and the optional rotating manipulator create pinch points and high-speed propellers that can lacerate skin or entrap hair, clothing, or tools.                                                            |  |
|                                                                                                       | Measures Currently in place to prevent risk of injury                                                                                                                                                                                                                                                                                        |                                                                                      |
|                                                                                                       | <ul style="list-style-type: none"> <li>• Thruster guards fitted; pre-dive propeller checks.</li> <li>• Software lockout prevents motor start until diagnostics pass.</li> <li>• Pilots briefed to keep hands clear and disable “Auto” modes before recovery.</li> <li>• Emergency stop on console; soft-stop in control software.</li> </ul> |                                                                                      |

| Potential Rating                                                                                        | Additional Controls Required                                                                                                                                                                     |
|---------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>High Risk</b><br> | <b>No Remedial Actions Entered</b><br>If there are any reasonably practicable actions which can be taken to reduce the risk associated with this hazard please use the form above to enter them. |

| Current Rating                                                                                              | Hazard Information                                                                                                                                                                                                                        | Task Status                                                                           |
|-------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| <b>Moderate Risk</b><br> |  <b>Entrapment/Entanglement</b><br>Floating tether and strain-relief cables can loop around limbs or equipment, posing drag-in, trip, or snag hazards. |  |
|                                                                                                             | Measures Currently in place to prevent risk of injury                                                                                                                                                                                     |                                                                                       |
|                                                                                                             | <ul style="list-style-type: none"> <li>• Mandatory strain-relief lanyards and carabiners; tether managed by a designated handler.</li> <li>• Clear deck markings and “no-step” zones around tether.</li> </ul>                            |                                                                                       |

| Potential Rating                                                                                            | Additional Controls Required                                                                                                                                                                     |
|-------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Moderate Risk</b><br> | <b>No Remedial Actions Entered</b><br>If there are any reasonably practicable actions which can be taken to reduce the risk associated with this hazard please use the form above to enter them. |

| Current Rating                                                           | Hazard Information                                                                                                                                                                                                                                                                                                      | Task Status                                     |
|--------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|
| Moderate Risk<br><div><div></div><div></div><div></div><div></div></div> | <div><div><div></div></div><div><b>Drowning / Asphyxiation</b><br/>Work is conducted at water's edge; personnel could fall in while launching/recovering the ROV or troubleshooting submerged hardware.</div></div>                                                                                                     | <div><div>0</div><div>0</div><div>0</div></div> |
|                                                                          | Measures Currently in place to prevent risk of injury                                                                                                                                                                                                                                                                   |                                                 |
|                                                                          | <ul style="list-style-type: none"><li>• Personal flotation devices (PFDs) available and worn when working over open water.</li><li>• Two-person rule during launch/recovery; guardrails or poolside barriers installed where practical.</li><li>• Rescue pole and throw-line stored within 5 m of test tanks.</li></ul> |                                                 |

| Potential Rating                                                         | Additional Controls Required                                                                                                                                                                                            |
|--------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Moderate Risk<br><div><div></div><div></div><div></div><div></div></div> | <div><div><b>No Remedial Actions Entered</b><br/>If there are any reasonably practicable actions which can be taken to reduce the risk associated with this hazard please use the form above to enter them.</div></div> |